International Islamic University Chittagong (IIUC)

Department of Computer and Communication Engineering

Midterm Examination

Program: **B.sc (Engg.)**
Course Code: **CCE-3511**
Total Marks: **30**

Semester: **Spring-2026**
Course Title: **Electromagnetic Fields and Waves**
Time: **1 Hours 30 Minutes**

(i) Answer all the questions. The figures in the right-hand margin indicate full marks.
(ii) Course Learning Outcomes (CLOs) and Bloom's Levels are mentioned in additional Columns.

Course Learning Outcomes (CLOs) of the Questions	
CLO1	Understand and analyze the basic concepts related to Electrostatics, Magnetostatics and Electromagnetic Waves.
CLO2	Determine Electric and Magnetic Field for different scenarios with varying specifications using different techniques.
CLO3	Analyze propagation of Electromagnetic Waves in different mediums using Maxwell's Equations.

Bloom's Levels of the Questions

Letter Symbols	**R**	**U**	**Ap**	**An**	**E**	**C**
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

Q1.	**a)**	Mention the differences between electric field and magnetic field. Deduce the relationship between relative permittivity and electric susceptibility.	**CLO1**	**U**	**5**
	b)	State and explain coulomb's law. What is electric dipole?	**CLO1**	**U**	**5**
Q2.	**a)**	State and explain Gauss's law. Derive the electric field of a point charge.	**CLO1**	**U**	**5**
	b)	The two plates of a parallel-plate capacitor are separated by a distance d and maintained at potentials 0 and V_0. Assuming negligible fringing effect at the edges, determine (i) the potential at any point between the plates, and (ii) the surface charge densities on the plates.	**CLO2**	**Ap**	**5**
Q3.	**a)**	Define the following terms: (i) Displacement current, (ii) Electric field intensity, (iii) Electric flux intensity, and (iv) Potential difference.	**CLO1**	**U**	**4**
	b)	Determine the electric field intensity at P(-0.2, 0, -2.3) due to point charge of + 5 nC at Q(0.2, 0.1, -2.5) in air. All dimensions are in meters.	**CLO2**	**Ap**	**6**

OR

Q3.	**a)**	What is meant by volume charge density? State and explain Biot-Savart's law.	**CLO1**	**U**	**4**
	b)	Starting from Gauss's law derive the formula of Poisson's and Laplace's equations. Prove that $\nabla \times \bar{H} = \bar{J}$; where the symbols have their usual meanings.	**CLO2**	**Ap**	**6**

$\oint E\,ds = \frac{Q}{\epsilon_0}$

International Islamic University Chittagong (IIUC)

Department of Computer and Communication Engineering

Midterm Examination

Program: **B.sc (Engg.)**	Semester: **Autumn-2025**
Course Code: **CCE-3511**	Course Title: **Electromagnetic Fields and Waves**
Total Marks: **30**	Time: **1 Hours 30 Minutes**

(i) Answer all the questions. The figures in the right-hand margin indicate full marks.
(ii) Course Learning Outcomes (CLOs) and Bloom's Levels are mentioned in additional Columns.

Course Learning Outcomes (CLOs) of the Questions

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CLO2	Determine Electric and Magnetic Field for different scenarios with varying specifications using different techniques.
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Bloom's Levels of the Questions

Letter Symbols	R	U	Ap	An	E	C
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

Q1.	**a)**	Define the following terms: (i) Electric field, (ii) Magnetic field, (iii) Displacement current, (iv) Electric field intensity, (v) Electric flux intensity, and (vi) Potential difference.	**CLO1**	**U**	**6**
	b)	What is meant by volume charge density? Deduce the relationship between relative permittivity and electric susceptibility.	**CLO1**	**U**	**4**
Q2.	**a)**	Derive the formula of Poisson's and Laplace's equations.	**CLO1**	**U**	**4**
	b)	State and explain Biot-Savart's law. Prove that $\nabla \times \bar{H} = \bar{J}$; where the symbols have their usual meanings.	**CLO2**	**Ap**	**6**
Q3.	**a)**	State and explain Gauss's law. Derive the electric field of a point charge.	**CLO1**	**U**	**5**
	b)	The two plates of a parallel-plate capacitor are separated by a distance d and maintained at potentials 0 and V_0. Assuming negligible fringing effect at the edges, determine (i) the potential at any point between the plates, and (ii) the surface charge densities on the plates.	**CLO2**	**Ap**	**5**

OR

Q3.	**a)**	State and explain coulomb's law. What is electric dipole?	**CLO1**	**U**	**5**
	b)	Determine the electric field intensity at P(-0.2, 0, -2.3) due to point charge of + 5 nC at Q(0.2, 0.1, -2.5) in air. All dimensions are in meters.	**CLO2**	**Ap**	**5**

International Islamic University Chittagong (IIUC)
Department of Computer and Communication Engineering
Midterm Examination

Program: **B.sc (Engg.)**
Course Code: **CCE-3511**
Total Marks: **30**

Semester: **Spring-2025**
Course Title: **Electromagnetic Fields and Waves**
Time: **1 Hours 30 Minutes**

(i) Answer all the questions. The figures in the right-hand margin indicate full marks.
(ii) Course Learning Outcomes (CLOs) and Bloom's Levels are mentioned in additional Columns.

Course Learning Outcomes (CLOs) of the Questions	
CLO1	Understand and analyze the basic concepts related to Electrostatics, Magnetostatics and Electromagnetic Waves.
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Bloom's Levels of the Questions						
Letter Symbols	**R**	**U**	**Ap**	**An**	**E**	**C**
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

Q1.	**a)**	What is electric dipole? State and explain coulomb's law.	**CLO1**	**U**	**5**
	b)	Mention the differences between electric field and magnetic field. Deduce the relationship between relative permittivity and electric susceptibility.	**CLO1**	**U**	**5**
Q2.	**a)**	State and explain Biot-Savart's law.	**CLO1**	**U**	**4**
	b)	Prove that $\nabla \times \vec{H} = \vec{J}$; where the symbols have their usual meanings. Starting from Gauss's law derive the formula of Poisson's and Laplace's equations.	**CLO2**	**Ap**	**6**
Q3.	**a)**	State and explain Gauss's law. Derive the electric field of a point charge.	**CLO1**	**U**	**5**
	b)	The two plates of a parallel-plate capacitor are separated by a distance d and maintained at potentials 0 and V_0. Assuming negligible fringing effect at the edges, determine (i) the potential at any point between the plates, and (ii) the surface charge densities on the plates.	**CLO2**	**Ap**	**5**

OR

Q3.	**a)**	What is meant by vector magnetic potential? Write the physical significance of vector magnetic potential.	**CLO1**	**U**	**4**
	b)	Determine the electric field intensity at P(-0.2, 0, -2.3) due to point charge of + 5 nC at Q(0.2, 0.1, -2.5) in air. All dimensions are in meters.	**CLO2**	**Ap**	**6**

International Islamic University Chittagong (IIUC)
Department of Computer and Communication Engineering
Midterm Examination

Program: **B.sc (Engg.)**
Course Code: **CCE-3511**
Total Marks: **30**

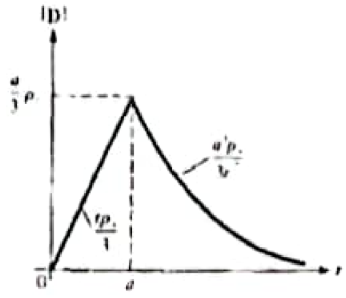
Semester: **Spring 2024**
Course Title: **Electromagnetic Fields and Waves**
Time: **1 Hour and 30 Minutes**

(i) Answer all the questions. The figures in the right-hand margin indicate full marks.
(ii) Course Outcomes (COs) and Bloom's Levels are mentioned in additional Columns.
'x'= Last two digits of Student ID

Course Outcomes (COs) of the Questions	
CLO1	Understand and analyze the basic concepts related to Electrostatics, Magnetostatics and Electromagnetic Waves.
CLO2	Determine Electric and Magnetic Field for different scenarios with varying specifications using different techniques.

Bloom's Levels of the Questions						
Letter Symbols	**R**	**U**	**Ap**	**An**	**E**	**C**
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

Q1.	**a)**	Deduce Coulomb's law from Gauss's law thereby affirming that Gauss's law is an alternative statement of Coulomb's law. [Hint: Determine **E** for Point charge using Gauss's law, then find **F**).	**CLO1**	**U**	**3**
	b)	Point charges Q_1 = 5 μC and Q_2 = -4 μC are placed at (3, x, 1) and (-2x, 0, 6) respectively. Determine the force on Q1.	**CLO2**	**Ap**	**4**
	c)	Draw the typical fields for the following four categories. i) $\nabla.\overline{A}= 0; \nabla \times \overline{A} = 0$ ii) $\nabla.\overline{A}\neq 0; \nabla \times \overline{A} = 0$ iii) $\nabla.\overline{A}= 0; \nabla \times \overline{A} \neq 0$ iv) $\nabla.\overline{A}\neq 0; \nabla \times \overline{A} \neq 0$	**CLO2**	**An**	**3**
		OR			
	a)	Point out for what condition of charge are Electrostatic and Magnetostatic fields created?	**CLO1**	**U**	**2**
	b)	Explain the statement:"Electrostatic Fields are Irrotational and Divergent".	**CLO1**	**Ap**	**3**
	c)	Given that **D** = $Z_\rho \cos^2\varphi\, a_z$ C/m², calculate the charge density at (1, π/4, 3) and the total charge enclosed by the cylinder of radius 1m with - 2≤ z ≤2 m.	**CLO2**	**Ap**	**5**
Q2.	**a)**	"An isolated magnetic charge does not exist"- Explain with the help of Gauss's law of Magnetostatics.	**CLO1**	**U**	**3**
	b)	Why is Amperian Path important? Derive Magnetic Field intensity for an infinite line current. What is the meaning of "Infinite"- here?	**CLO2**	**Ap**	**5**

	c)	Write the Differential and Integral form of Gauss's Law and Ampere's Law for Magnetostatic Field.	CLO1	R	2
Q3.	a)	Why are Electrostatic Boundary value problems important?	CLO1	U	2
	b)	Derive Poisson's Equation for Electrostatic Potential from Gauss's Law.	CLO1	U	3
	c)	Explain the prove the following graph in Fig. 1 for uniformly charged sphere, where symbols carry their usual meaning.  Fig. 1	CLO2	An	5

International Islamic University Chittagong (IIUC)

Department of Computer and Communication Engineering

Midterm Examination

Program: **B.sc (Engg.)**
Course Code: **CCE-3511**
Total Marks: **30**

Semester: **Spring 2022**
Course Title: **Electromagnetic Fields and Waves**
Time: **1 Hour 30 Minutes**

(i) Answer all the questions. The figures in the right-hand margin indicate full marks.
(ii) Course Learning Outcomes (CLOs) and Bloom's Levels are mentioned in additional Columns.

Course Learning Outcomes (CLOs) of the Questions

CLO1	Understand and analyze the basic concepts related to Electrostatics, Magneto statics and Electromagnetic Waves.
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Bloom's Levels of the Questions

Letter Symbols	**R**	**U**	**Ap**	**An**	**E**	**C**
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

Q1.	a)	State Coulomb's law. A charge of $\mathbf{Q1 = -10\mu C}$ is placed at the origin of a rectangular coordinate system and a second charge, $\mathbf{Q2 = -10mC}$ is placed on the x-axis at a distance of **50 cm** from the origin. Find the force on Q1 due to Q2 if they are in free space.	CLO1	R+ Ap	2+3
	b)	For an infinitely charged sheet, prove that field due to surface charge density $$E = \frac{\rho_s}{2\epsilon_0} a_n$$	CLO1	An	5
Q2.	a)	Derive the equation for potential at a point due to a point charge.	CLO1	Ap	4
	b)	Show that, ptential difference between two poits A and B is, $$V_{AB} = V_A - V_B$$	CLO1	An	3
	c)	If the potential function, V is given by $V = x^3y - xy^2 + 3z$, find the potential gradient.	CLO1	Ap	3
Q3.	a)	State Gauss's Law and prove it for a spherical surface which encloses a charge Q at its centre.	CLO1	An	5
	b)	From the point form of Gauss's Law, derive Poisson's and Laplace's equation	CLO1	Ap	5
		OR			
Q3.	a)	Write down the boundary conditions on **E** and D and prove the conditions foe two different media.	CLO1	An	10

Department of Computer and Communication Engineering

Midterm Examination

Program: B.sc (Engg.)
Course Code: CCE-3511
Total Marks: 30

Semester: Spring 2(
Course Title: Electromagnetic Fields and Wa
Time: 1 Hours 30 Minu

(i) Answer all the questions. The figures in the right-hand margin indicate full marks.
(ii) Course Outcomes (COs) and Bloom's Levels are mentioned in additional Columns.

Course Outcomes (COs) of the Questions

CLO1	Understand and analyze the basic concepts related to Electrostatics, Magnetostatics and Electromagnetic Waves.
CLO2	Determine Electric and Magnetic Field for different scenarios with varying specifications using differe techniques.
CLO3	Analyze propagation of Electromagnetic Waves in different mediums using Maxwell's Equations.

Bloom's Levels of the Questions

Letter Symbols	R	U	Ap	An	E	C
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Cre

Q1.	**a)**	State Coulomb's law. A charge of Q1 = $-1\cdot0$ μC is placed at the origin of a rectangular coordinate system and a second charge, Q2 = -10 mC is placed on the x-axis at a distance of 50 cm from the origin. Find the force on Q1 due to Q2 if they are in free space.	CLO1	R,Ap
	b)	For an infinitely charged sheet, prove that field due to surface charge density $$E = \frac{\rho_s}{2\epsilon_0} a_n$$	CLO1	An
Q2.	**a)**	Derive the equation for potential at a point due to a point charge.	CLO1	A
	b)	Show that, ptential difference between two poits A and B is, $V_{AB} = V_A - V_B$	CLO1	A
	c)	If the potential function, V is given by $V = x^3y - xy^2 + 3z$, find the potential gradient.	CLO1	
Q3.	**a)**	Define electric flux and electric flux density. If an electric field in free space is given by, $\mathbf{E} = a_x + 2a_y + 5a_z$ V/m, find the electric flux density.	CLO1	E
	b)	State Gauss's Law and prove it for a spherical surface which encloses a charge Q at its centre.	CLO1	

OR

Q3.	**a)**	"The tangential component of **E** is continuous across any boundary"- justify this statement.	CLO1
	b)	For normal component of **D**, prove that $D_{n1} - D_{n2} = 0$	CLO1